Title: Introduction to Machine Learning Systems

Machine learning systems are designed to learn patterns from data and make predictions or decisions without being explicitly programmed. These systems are widely used in applications such as recommendation engines, fraud detection, healthcare diagnostics, and natural language processing.

A typical machine learning pipeline consists of several stages. The first stage is data collection, where raw data is gathered from various sources such as databases, APIs, or user interactions. The second stage is data preprocessing, which involves cleaning the data, handling missing values, normalizing features, and removing noise.

Feature engineering is another critical step in the pipeline. In this stage, raw data is transformed into meaningful features that can improve model performance. For example, in a customer churn prediction system, features such as customer tenure, monthly charges, and service usage patterns are often used.

After preprocessing and feature engineering, the data is split into training and testing sets. The model is trained on the training data and evaluated on the testing data. Common evaluation metrics include accuracy, precision, recall, and F1-score.

Section: Model Deployment

Once a model is trained and evaluated, it is deployed into a production environment. Deployment allows the model to make predictions on real-world data. This is often done through APIs, where applications can send requests and receive predictions in real time.

However, deploying a model is not the end of the process. In production, models must be monitored continuously to ensure they perform as expected. Over time, the data distribution may change, leading to a decline in model performance.

Section: Data Drift and Concept Drift

Data drift occurs when the statistical properties of input data change over time. For example, if a model was trained on customer data from 2022, but the current data is from 2025, the distribution of features such as income or purchasing behavior may have shifted.

Concept drift occurs when the relationship between input features and the target variable changes. For instance, a feature that was once a strong predictor of customer churn may no longer be relevant due to changes in business strategy or customer behavior.

Handling drift typically involves retraining the model with updated data, adjusting features, or modifying thresholds used for decision-making.

Section: Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a technique used to improve the performance of language models by combining retrieval and generation. Instead of relying solely on the model’s internal knowledge, RAG retrieves relevant documents from a knowledge base and uses them as context for generating answers.

A RAG system typically consists of a document store, an embedding model, a vector database, and a language model. When a user submits a query, the system converts the query into an embedding and retrieves the most relevant documents based on similarity.

These retrieved documents are then passed to the language model along with the query. The model generates a response that is grounded in the retrieved context, reducing hallucinations and

improving accuracy.

Section: Challenges in RAG Systems

Despite its advantages, RAG systems come with challenges. One major challenge is selecting the right chunk size for documents. If chunks are too small, important context may be lost. If chunks are too large, retrieval may become less precise.

Another challenge is ensuring high-quality embeddings. Poor embeddings can lead to irrelevant document retrieval, which negatively impacts the generated response.

Latency is also a concern, especially in real-time applications. Efficient indexing and retrieval mechanisms are required to maintain fast response times.